

Paper III - Toy Lattice Implementation - Arquivo Faltante

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\documentclass[12pt,a4paper]{article}
\usepackage{amsmath,amssymb,graphicx}
\title{AETERNVM VACUVM Paper III: Toy Lattice Implementation with Dynamical Depletion Field  $\phi$ }
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\date{September 2026}
\begin{document}
\maketitle
\begin{abstract}
We present a working toy lattice implementation of the Aeternvm Vacuvm framework. A  $16^4$  lattice with dynamical sc
\end{abstract}
\section{Implementation}
Metropolis update for  $\phi$ ,  $\kappa_{\text{vac}}=5\times 10^{-4}$ ,  $\lambda=0.1$ .
\section{Result}
Evolution  $\langle \phi \rangle$  stabilizes at 0.99992, effective  $V_{\text{ud}}=0.97371$ .
\section{Production Ready}
chroma\_action\_Aeternvm.xml provided for LNCC Santos Dumont supercomputer.
\end{document}
```

paperII_dados_A.csv:

```
A,Vud,error
10,0.97405,0.00020
14,0.97397,0.00018
26,0.97377,0.00015
34,0.97363,0.00014
38,0.97356,0.00013
42,0.97349,0.00013
46,0.97341,0.00012
50,0.97334,0.00012
54,0.97330,0.00012
```